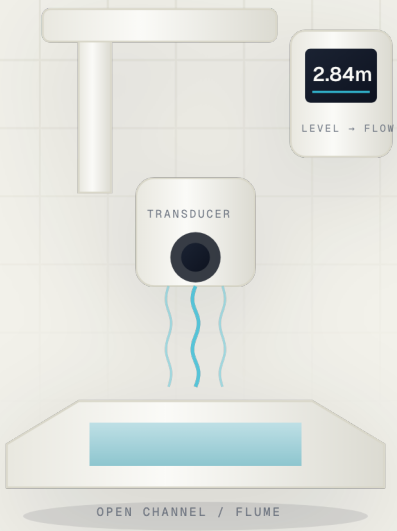
VEPOLINK
MONITORING PURITY

PROCESS INSTRUMENTATION × FLOW MEASUREMENT

NON-CONTACT LEVEL-TO-FLOW FOR OPEN CHANNELS

EchoFlow Transmitter *Non-Contact Ultrasonic*

An airborne ultrasonic level transmitter that converts head level at a flume or weir into flow — mounted clear of the liquid, with no wetted electronics and no moving parts.



0.25–12 m
LEVEL MEASURING RANGE

±0.15 % / 6mm
ACCURACY, WHICHEVER IS GREATER

3 mm
RESOLUTION & REPEATABILITY

PULSE-ECHO TIME-OF-FLIGHT | **AUTO** FALSE-ECHO SUPPRESSION | **VEPOLINK** FIELD-PROVEN BUILD

12m	3mm	0contact	IP68
MAXIMUM LEVEL RANGE	RESOLUTION, REPEATABILITY	NO WETTED ELECTRONICS	SUBMERSION-RATED ENCLOSURE

01 OVERVIEW

Flow, measured *from the air above.*

A non-contact ultrasonic level transmitter mounted above a wet well, flume or weir, measuring level by airborne pulse-echo and converting it to flow using a standard head-discharge equation. Nothing touches the liquid, so there is nothing to foul, corrode or wear in continuous service.

Auto false-echo suppression maps out fixed obstructions such as ladders, pipes and turbulence sources, so the true liquid surface echo is tracked reliably even in a crowded channel.

REPORTED CHANNELS

- LIQUID LEVEL
- COMPUTED FLOW RATE
- TOTALISED VOLUME
- 4-20 MA OUTPUT
- HART COMMUNICATION
- PROFIBUS PA OPTION
- SIGNAL STRENGTH / ECHO CONFIDENCE
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

Non-Contact Sensing

The transducer sits above the channel, measuring level by airborne ultrasonic pulse with nothing touching the liquid.

Auto False-Echo Suppression

Fixed obstructions in the channel are mapped and ignored automatically, keeping lock on the true liquid surface.

Level-to-Flow Conversion

Built-in head-discharge equations convert measured level at a flume or weir directly into a flow reading.

HART / PROFIBUS PA

Standard 4–20 mA with HART, or PROFIBUS PA, integrates the transmitter into most existing plant control systems.

02 MEASUREMENT PRINCIPLE

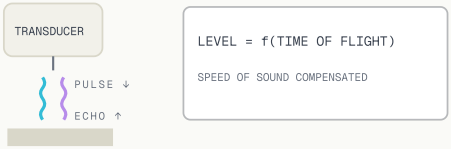
How an echo *becomes a flow rate.*

An ultrasonic pulse travels down to the liquid surface and back; the time of flight gives level. That level, measured at a calibrated flume or weir, converts directly to flow through a known head-discharge relationship.

TIME OF FLIGHT

Pulse-Echo Ranging

AIRBORNE PULSE · ECHO RETURN · SPEED-OF-SOUND COMPENSATION



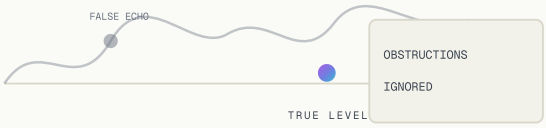
The transducer emits an ultrasonic pulse toward the liquid surface and times its return. Level is computed from that time of flight, compensated for the speed of sound in air.

`Level = f(time of flight)`

SIGNAL PROCESSING

Auto False-Echo Suppression

OBSTRUCTION MAPPING · ECHO CONFIDENCE



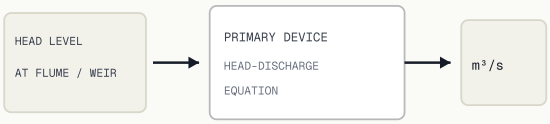
Fixed obstructions such as brackets, ladders or turbulence sources produce their own echoes; the transmitter learns and ignores these, tracking only the true liquid-surface return.

`Reliable lock in a crowded channel`

CONVERSION

Level-to-Flow

FLUME / WEIR · HEAD-DISCHARGE EQUATION



At a calibrated primary device — a Parshall flume, V-notch or rectangular weir — measured head converts directly to flow using a known head-discharge equation.

`Head level → m³/s`

OUTPUT

Display & Communication

4-20 MA · HART · PROFIBUS PA



Level, flow and totalised volume are shown on a local LCD with bar graph, and carried onward over loop current, HART or PROFIBUS PA to the plant system.

`Local display · remote reporting`

03 DEPLOYMENT

From channel *to control room.*

Everything from the transducer mount to the control-room reading is chosen to keep the measurement stable in an open, often turbulent, outdoor environment.

STEP 01

Flange mount over channel

A universal 80 mm flange mounts the transducer directly above the flume, weir or wet well, clear of the liquid surface.

STEP 02

Pulse-echo ranging

Ultrasonic signal processing tracks the true liquid echo through turbulence, foam and vapour.

STEP 03

Level-to-flow conversion

Programmed head-discharge curves for the site's flume or weir turn the level reading into a flow rate.

STEP 04

Report & log

Flow and totalised volume are displayed locally and transmitted over loop, HART or PROFIBUS PA.

CHEMICAL COMPATIBILITY WITHOUT CONTACT

Because the transducer never touches the liquid, chemical compatibility is set by the transducer face material — ETFE or PVDF — rather than the whole wetted assembly, simplifying selection for aggressive or variable chemistries.

PROGRAMMING & CONFIGURATION

Configuration is done via an intrinsically safe infrared handheld programmer, a PC-based device manager, or a HART communicator — no need to open the enclosure in the field.

WHY THIS MEASUREMENT MATTERS

Open-channel flow is the standard way plants measure inlet and outlet flow at treatment works, irrigation offtakes and stormwater discharges — any site where water runs in a channel rather than a closed pipe. A reliable, non-contact level reading is what keeps that flow record defensible.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

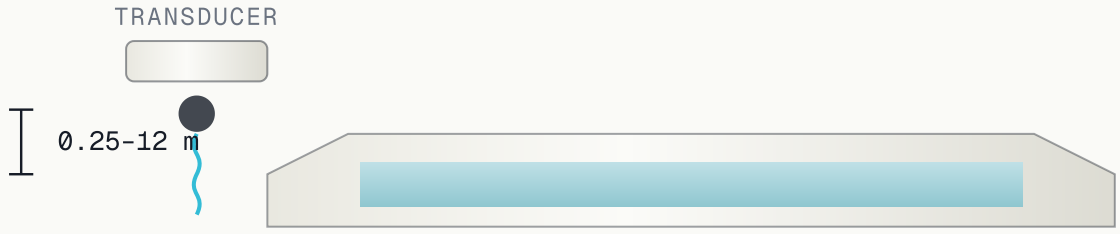
Parameter	Level, converted to open-channel flow
Working Principle	Non-contact ultrasonic pulse-echo
Range	0.25 - 12 m
Resolution	3 mm
Accuracy	± the greater of 0.15% of range or 6 mm
Repeatability	3 mm
Working Pressure	0.5 bar g
Working Temperature	-40 to 85 °C

SYSTEM · BUILD · POWER

Display	Local LCD with bar graph; remote via HART / PROFIBUS PA
Configuration	Infrared handheld programmer · PC device manager · HART communicator
Wetted Parts	ETFE or PVDF transducer face
Enclosure Material	PBT (polybutylene terephthalate)
Transducer Seal	Buna-N seal, ETFE or PVDF face
Output	4-20 mA · HART · PROFIBUS PA
Power Supply	Loop-powered, nominal 24 VDC (4-20 mA / HART) or PA bus powered
Protection Class	Type 4X / 6 · IP67 / IP68 enclosure
Installation	Non-contact · 3 inch (80 mm) universal flange
Remote Cable Length	As required by site layout

MOUNTING GEOMETRY

TRANSDUCER ABOVE CHANNEL · INDICATIVE, NOT TO SCALE



05 INSTALLATION & UPKEEP

Mounted above, *left running.*

The transducer mounts on a bracket above the flume, weir or wet well on its universal **80 mm flange**, aimed straight down at the liquid surface with clearance from walls and obstructions within its beam angle.

01 · MOUNTING

Centered, vertical

Mount centered over the channel and vertical to the liquid surface, clear of inflow turbulence and foam-prone zones.

02 · BEAM CLEARANCE

Away from obstructions

Keep brackets, ladders and pipework outside the beam angle where practical; auto false-echo suppression handles the rest.

03 · PRIMARY DEVICE

Calibrated flume or weir

Level-to-flow accuracy depends on a correctly installed, calibrated flume or weir upstream of the measurement point.

04 · CONFIGURATION

Programmed on site

Site-specific head-discharge parameters are set using the infrared programmer, PC tool or HART communicator at commissioning.

05 · POWER & SIGNAL

Loop or bus powered

Standard 4–20 mA/HART loop power or PROFIBUS PA bus power, run back to the control room or local panel.

06 · MAINTENANCE

Minimal, non-contact

With no wetted electronics and no moving parts, upkeep is limited to periodic cleaning of the transducer face and a level verification check.

SELECTING A PRIMARY DEVICE

Flow range, channel geometry and expected debris or solids content determine whether a Parshall flume, V-notch weir or rectangular weir suits the site. The transmitter's head-discharge conversion is configured to match whichever primary device is installed.

COMMISSIONING

Commissioning covers a dry and wet level verification against a staff gauge, confirming the beam locks on the true surface and not a false echo, and checking the programmed head-discharge curve against the installed primary device before handover.

06 APPLICATIONS

Where it *works.*

Any open channel, wet well or storage vessel where level or level-derived flow needs measuring without anything touching the liquid.

Water & Wastewater Treatment	Chemical Storage Vessels	Small Bulk Hoppers
Filter Beds	Liquid Storage Vessels	Irrigation & Canal Offtakes
Stormwater Monitoring	Industrial Discharge Points	

WHERE NON-CONTACT MATTERS MOST

Liquids carrying grit, grease, fibrous solids or aggressive chemistry foul or damage a wetted sensor over time. Measuring from above the surface removes that failure mode entirely, trading only beam-path clearance for a sensor that never touches what it measures.

LEVEL OR FLOW, ONE INSTRUMENT

The same transmitter serves pure level monitoring in a storage vessel or hopper, or level-to-flow conversion at a flume or weir — the configuration, not the hardware, decides which.

07 FAQ

Frequently asked *questions.*

What is the working principle?

Non-contact ultrasonic pulse-echo. A transducer mounted above the liquid times a pulse's round trip to compute level, which can then be converted to flow.

How does level become a flow reading?

At a calibrated flume or weir, measured head level converts to flow through a known head-discharge equation, programmed into the transmitter.

Does anything touch the liquid?

No. Measurement is entirely airborne — the transducer face sits above the surface, with no wetted electronics.

What is the measuring range?

0.25 to 12 metres, with 3 mm resolution and an accuracy of the greater of 0.15% of range or 6 mm.

How does it handle obstructions in the channel?

Auto false-echo suppression learns and ignores fixed obstructions such as brackets or ladders, keeping lock on the true liquid surface.

What outputs are available?

Standard 4–20 mA with HART, or PROFIBUS PA, plus a local LCD display with bar graph.

How is it configured?

Via an intrinsically safe infrared handheld programmer, a PC-based device manager, or a HART communicator — no enclosure access needed.

Is it suitable for aggressive liquids?

Yes. Since only the transducer face is exposed, chemical compatibility is set by the ETFE or PVDF face material rather than a fully wetted sensor body.

Can it be installed outdoors?

Yes, the enclosure is rated Type 4X/6 and IP67/IP68, suited to outdoor channels and wet wells across -40 to 85 °C.

What maintenance does it need?

Very little — periodic cleaning of the transducer face and a level verification check against a staff gauge. There are no moving parts or consumables.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

FLOW, MEASURED WITHOUT CONTACT

Know your flow. *From above.*

Tell us the channel, the primary device and the range you expect, and we'll configure the transducer, mount and output around your site. Pick how you'd like to start.

01

Talk to our engineers

Speak directly to the team that sizes, specifies and commissions open-channel flow transmitters. No call centre.

+91 99538 86178 →

02

Request a sizing & configuration

RECOMMENDED

Send your channel, primary device and expected range — we'll spec the transducer, mount and output.

sales@vepolink.com →

03

Book a demonstration

See a running transmitter and the level-to-flow output with our engineering team.

Request a demo →